



**National College of Ireland**  
**Project Submission Sheet**

|                             |                                                                  |              |      |
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I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

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| <b>Date:</b>      | 03/05/2026                                                       |

# Dublin Housing Affordability Analysis (2015–2025)

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**Abstract**—Policy discussions on Dublin housing often focus on sale prices alone. Our group argues that affordability depends on how prices move relative to earnings, not on nominal prices by themselves. Between 2015 and 2025 we asked whether Dublin housing grew harder to afford compared with income, and whether poor affordability clusters in particular parts of the city rather than spreading evenly.

We built an analytics pipeline in Python that pulls together three official Irish sources: Property Price Register transactions (CSV), CSO income series via JSON-stat APIs, and optional CSO population data for context. Raw extracts stay in MongoDB so we can trace what entered the system; cleaned yearly tables and area summaries load into PostgreSQL for querying and for a Streamlit dashboard alongside figures exported from static analysis.

Across the decade, average Dublin transaction prices rose sharply in our build. Where income series overlap our window, the ratio of average price to income moves against affordability—price gains exceed income gains. Area rankings and heatmaps suggest lasting gaps between locations, not one-off yearly noise. We summarise integration obstacles (API shape drift, sparse Eircodes, year alignment, relational typing) and the mitigations coded in the repository, and we point to natural extensions such as inflation adjustment and official small-area geography.

## I. INTRODUCTION

Rising sale prices in Dublin are widely reported, but affordability depends on prices relative to income, not on nominal levels alone. We study whether housing became harder to afford in that sense between 2015 and 2025, and whether high cost concentrates in some areas more than others.

The research question is:

*Is housing in Dublin becoming unaffordable relative to income over time (2015–2025), and how does affordability vary across areas?*

The module requires multiple related datasets (including semi-structured data), storage in a non-relational and a relational database, programmatic ETL, and static plus interactive visuals. Our design uses PPR CSV, CSO JSON-stat APIs, MongoDB for raw lineage, PostgreSQL for curated tables, and Python with Streamlit for the dashboard. Team members each led one stream (PPR extract, income API, visualization) while contributing across all stages.

Because PPR fields can include addresses, we treat extracts as sensitive in storage even though the source is a public register: repository hygiene (no secrets in Git, local databases not exposed) follows standard analytics practice and module expectations.

The formal objectives are:

- 1) Build a reproducible ETL pipeline across structured and semi-structured datasets.

- 2) Store raw and processed datasets programmatically in two database systems: MongoDB and PostgreSQL.
- 3) Engineer affordability indicators and area-level comparisons for Dublin.
- 4) Produce static analysis visuals and an interactive dashboard.
- 5) Document methods, challenges, findings, and limitations in IEEE-style format.

Scope is Dublin transactions and annual indicators for 2015–2025. We deliver a transparent descriptive pipeline—cleaning rules, joins, and outputs a reader can trace in code—and we separate findings from stronger claims that would need causal identification or richer microdata.

The novelty for this module is primarily *engineering and integration*: we combine high-volume administrative sales records with survey-derived income cubes under explicit join assumptions, retain raw payloads for audit, and surface the same metrics in both static IEEE figures and an interactive dashboard. That answers the research question at descriptive level; it does not identify causal drivers of price growth. For stakeholders, the value is an inspectable evidence chain from official URLs to charts, which is harder to achieve when analysis lives only in ad hoc spreadsheets.

## II. RELATED WORK

Affordability is operationalised in many ways: rent-to-income caps, mortgage stress tests, residual income after housing costs, and simple price-to-earnings ratios. Irish empirical discussion by Corrigan et al. [1] emphasises slow supply response and persistent demand, which helps interpret multi-year Dublin price drift. Norris and Byrne [2] tie affordability to tenure and welfare design; their lens cautions that a single national income series may not represent every household segment. We cite these works for motivation, not to replicate their estimators.

Internationally, OECD analysis [3] shows that affordability pressure can increase while GDP or wages grow if new housing does not expand elastically. Glaeser and Gyourko [4] provide a supply-constraint narrative in which prices can persistently outrun wages in regulated or land-scarce cities. Dublin differs in institutions from their US examples, but the ratio-based question we ask is in the same family: are asset prices decoupling from labour income?

For data, CSO PxStat [5] and ESRI synthesis [6] are authoritative entry points. The engineering gap is operational: PxStat returns JSON-stat cubes with dimension indices, whereas the Property Price Register is a wide CSV with string dates and

occasional missing geocodes. Student projects often gloss that join; ours documents it.

Synthesis reports are indispensable for context, yet they rarely ship machine-actionable join keys to row-level sales data. That limitation pushes applied coursework toward the PxStat API and open registers even when the narrative framing comes from higher-level studies. Our related-work use of ESRI and OECD sources is therefore interpretive: they motivate questions about distribution and supply, while the empirical series in this report are built from micro exports we actually parse.

#### A. Critical Evaluation of Prior Approaches

We see three recurring gaps in work adjacent to ours: (1) results without a reproducible path from URL to final table; (2) semi-structured sources flattened once with no retained payload for re-parse; (3) charts whose numbers cannot be tied to versioned transformation code. Our design responds with explicit modules, `data/raw` snapshots, MongoDB payload storage, PostgreSQL serving layers, and figures generated from the same columns as the Streamlit application.

Corrigan et al. [1] and Norris and Byrne [2] foreground supply sluggishness, credit cycles, and tenure shifts in Ireland; their value for us is framing rather than a plug-in empirical prior. Those studies typically work at national or coarse regional scales with survey or model-based income, whereas our exercise is deliberately granular on Dublin transactions and coarse on income alignment. A limitation of leaning on this literature is that our affordability index is a mechanical ratio of means, not a welfare measure or residual income test, so we borrow interpretive language cautiously and keep claims descriptive.

### III. DATA PROCESSING METHODOLOGY

#### A. Dataset Selection and Justification

We deliberately avoided Kaggle and similar mirrors where prefabricated notebooks encourage copy-paste solutions. All sources are Irish public-sector or government-backed, with stable landing pages suitable for citation in a methods section.

- 1) **Property Price Register (PPR)** [7]—<https://www.propertypriceregister.ie>: open CSV exports of residential transactions. We use Dublin rows as the numerator side of affordability (mean and median prices, counts). Volume is large enough to stress memory and typing if handled naively, which is intentional for a programming module.
- 2) **CSO income cube (SIA14, JSON-stat 2.0)** [5]—accessed via the PxStat REST pattern documented on <https://www.cso.ie>. This is semi-structured: dimensions are not fixed column names until parsed.
- 3) **CSO population cube (F1015)** [5]: optional context for a “price per thousand residents” style intensity measure; same API family as income.

Table I gives approximate scale; each source exceeds 1,000 records.

TABLE I  
APPROXIMATE RECORD SCALE AFTER INGESTION (ILLUSTRATIVE OF PROJECT BUILD).

| Source                   | Format            | Rows (approx.) |
|--------------------------|-------------------|----------------|
| PPR (Dublin subset)      | CSV               | 198,057        |
| Income (parsed cube)     | JSON-stat → table | 1,056          |
| Population (parsed cube) | JSON-stat → table | 13,020         |

#### B. Pipeline Architecture and Data-Flow Diagram

The PPR extract is large in row count; CSO cubes are smaller but structurally complex—a useful contrast for LO2 (“big” tabular volume versus semi-structured parsing cost on a laptop). Responsibilities are split so each file has a single reason to change: `extract.py` performs HTTP GET for CSO JSON, reads the PPR CSV, flattens JSON-stat into rows, writes MongoDB collections, and may snapshot `data/raw`; `transform.py` applies cleaning, merges income and optional population, builds yearly and area aggregates; `load.py` ensures DDL and reloads Postgres tables idempotently; `main.py` loads environment variables via `python-dotenv`, sequences steps, and can continue if Postgres is temporarily unavailable; `analyze.py` writes PNGs to `docs/figures`; `dashboard/app.py` reads processed CSVs for the interactive demo.

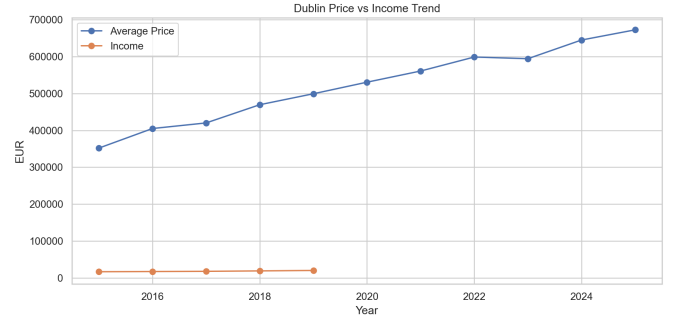


Fig. 1. City-level trend chart used in static and dashboard outputs.

Flow: Extract → MongoDB (raw) → Transform → PostgreSQL (curated) → figures and dashboard. Figure 1 illustrates a downstream trend output.

#### C. Reproducibility, Validation, and Operational Checks

Reproducibility is treated as a pipeline property, not a slogan. Each run can start from the same entry script; configuration points (database URIs, file paths) are externalised so a teammate can rebuild tables on a fresh machine after cloning the repository. During extraction we log HTTP status codes and write or refresh raw JSON alongside Mongo inserts when caching is enabled, which makes it possible to compare today’s download with yesterday’s if PxStat labels shift.

Validation is layered: schema-level checks after parsing (expected columns, non-null years for income rows), sanity bounds on prices, and spot comparisons of aggregate counts (Dublin rows before and after filters) printed to the console.

We do not claim statistical inference on duplicates in PPR beyond dropping impossible values; gifts and non-market transfers may still influence means, which is why medians are retained in parallel and why limitations are stated explicitly. This subsection documents behaviour a marker can trigger by re-running the pipeline rather than trusting a one-off export CSV.

County filtering uses the PPR county text field with a Dublin substring test, which matches the administrative County Dublin convention in the register. That includes Dublin city and county as reported by vendors, but it is not a travel-to-work or functional urban area boundary; comparisons with CSO regional labels therefore require accepting a degree of geographic mismatch. Documenting that mismatch is part of honest integration work, because a perfect boundary would need geocoding every row against official polygons—a sensible extension but outside our current scope.

#### D. Raw Data Persistence in MongoDB

Collections include `property_prices_raw` (row documents), `cso_income_raw` plus `cso_income_parsed_raw`, and the analogous population pair. Keeping both payload and parsed tables supports debugging when CSO revises labels: we can diff payloads and re-run the parser without guessing what changed. A purely relational landing zone would require either a wide “JSON blob” column in Postgres—possible but less natural than Mongo for rapid iteration—or lossy early flattening we wished to avoid.

#### E. Algorithms for Cleaning, Validation, and Transformation

1) *Property data:* Cleaning is deliberately rule-based (not ML) so examiners can audit thresholds. Steps: harmonise headers including alternate euro symbols; parse `sale_date` with Irish day-first order; strip currency symbols and thousands separators from prices; coerce to numeric and drop non-finite values; retain rows whose county field contains “Dublin”; restrict years to [2015,2025]; remove very low prices (e.g. below EUR 10,000) that often reflect gifts or data quirks. Each rule is visible in `transform.py` and can be debated or tuned.

2) *JSON-stat parsing:* The parser walks `dataset.id` for dimension order, expands category combinations from `dataset.dimension`, and zips them to `dataset.value`. If the API nests the cube under a top-level `dataset` key or declares `class=dataset`, we unwrap first—a branch added after a failed run on a cached payload with a different envelope.

3) *Income and population normalization:* We locate time columns by substrings such as `TLIST` or `TIME` and region-like columns by `REGION` or similar. Years and values are coerced to numeric, filtered to the analysis window, and averaged across duplicate years if the cube returns multiple Dublin-compatible rows. If strict Dublin/State/Ireland filtering would return an empty frame, we log a warning and fall back to all rows so a silent empty merge cannot masquerade as “no problem.”

#### F. Feature Engineering and Aggregation

Yearly metrics computed on the cleaned property frame:

- `avg_price_eur`, `median_price_eur`, `sales_count`.
- Merge with income on year (left join from prices so missing income years remain visible).
- `affordability_index` = `avg_price_eur / income_eur` when income is non-null.
- Year-over-year percentage changes for price, income, and the index where defined.

Optional `price_per_1000_residents` when population joins. Area = Eircode prefix if present, else address parsing (Dublin district or locality); groups with fewer than ten sales are dropped for stability (limitation: less coverage). The affordability index is intentionally omitted wherever income is missing: we do not forward-fill or interpolate a denominator, because that would state a ratio the source tables do not support.

#### G. Processed Storage in PostgreSQL

Tables: `dublin_affordability_yearly`, `dublin_affordability_area`, `project_insights`. Reloads use `truncate/delete-then-append`. `area` was widened from `VARCHAR(20)` to `VARCHAR(120)` after long address-derived labels caused insert errors.

#### H. Technology Justification

Python with `pandas/numpy`; `requests`; `pymongo`; `sqlalchemy/psycopg2`; `streamlit/plotly`. The stack matches teaching norms and separates flexible raw storage from relational serving.

Configuration uses `python-dotenv` so credentials stay out of Git; idempotent Postgres reloads (clear then append) recover cleanly from failed runs at the cost of brief table unavailability, acceptable for batch coursework but not for live OLTP.

#### I. Programming and Data Processing Challenges

The brief asks for obstacles and remedies we actually encountered:

- 1) **Packaging scripts:** In PowerShell, `param(...)` must be the first executable statement; an early `$ErrorActionPreference` line caused “param is not recognised” errors until reordered.
- 2) **Area labels:** Relying only on Eircode prefixes left most rows in a single “unknown” bucket because many PPR rows lack Eircodes; we added address-based fallback (Dublin district patterns and locality tokens) and filtered small-*n* groups.
- 3) **Misleading visuals:** Plotting affordability through 2025 while income ended earlier implied a false flat tail; the dashboard now restricts the affordability series to years with both numerator and denominator.

- 4) **Database schema:** PostgreSQL rejected inserts when area exceeded VARCHAR(20); we widened to VARCHAR(120) and added an ALTER in DDL setup.
- 5) **API variability:** CSO region labels did not always match our initial filter strings; fallback logic and logging prevented brittle failures.

#### IV. DATA VISUALIZATION METHODOLOGY

##### A. Design Principles

We followed guidance on reducing chartjunk and matching encodings to tasks [8], [9]. Cleveland and McGill’s ordering of graphical perception ranks position along a common scale above area, colour hue, and volume [8]; that motivated keeping city-level price and income on one line chart with a shared vertical axis instead of a dual-axis presentation that can visually suggest correlation. For matrix views, sequential colour with monotonic luminance (e.g. Viridis) supports magnitude comparisons and reduces the risk of rainbow ramps that are not perceptually ordered. We also avoid redundant encodings: when bar colour previously mapped transaction counts while height mapped price, viewers had to decode two competing channels for one comparison task; counts now appear in hover detail only.

Gestalt grouping informed dashboard layout: related time-series sit adjacent, spatial rankings follow city trends, and text insights sit last so numeric context is established before qualitative bullets. The intent is to reduce extraneous cognitive load during both live demo and static reading of exported figures.

##### B. Chart Selection Rationale

- Lines for city price vs. income (single panel, shared axis; no dual-axis distortion).
- Line for affordability index, clipped to years with income so the series is not extrapolated.
- Bars to rank areas by mean price for a selected year.
- Heatmap for area $\times$ year means, restricted to top areas for legibility.

##### C. Color, Accessibility, and Interaction

Blues for baseline series, red accent for stress, sequential heatmap scales. Streamlit uses a year slider and hovers; limited widgets keep the demo stable.

Plotly inside Streamlit gives responsive charts and hovers with little boilerplate; static PNGs from `analyze.py` keep the IEEE PDF light. A single year selector limits widget state bugs during demo.

##### D. Dashboard Composition

Order: KPIs  $\rightarrow$  city trends  $\rightarrow$  affordability  $\rightarrow$  area bars  $\rightarrow$  heatmap  $\rightarrow$  text insights (overview, then detail).

#### V. RESULTS AND EVALUATION

##### A. Primary Findings

Mean Dublin transaction prices rise across 2015–2025 in our processed series. When CSO income aligns on the yearly merge, the affordability index—mean price divided by mean income—moves upward: the average dwelling, measured this way, requires more years of that income stream than at the start of the overlap. We stress that the mean is a summary statistic; median buyers, first-time buyers with caps, and cash purchasers face different effective prices. Still, for coursework the index gives a consistent apples-to-apples comparison across years under documented rules.

Dashboard KPIs restate the latest available yearly aggregates so a viewer can reconcile headline numbers with the line charts; any discrepancy would indicate a filter mismatch rather than an intentional story beat. We report direction and relative movement of the ratio rather than point estimates suitable for policy thresholds, because the income side is an annual macro-compatible series, not a matched cohort of buyers.

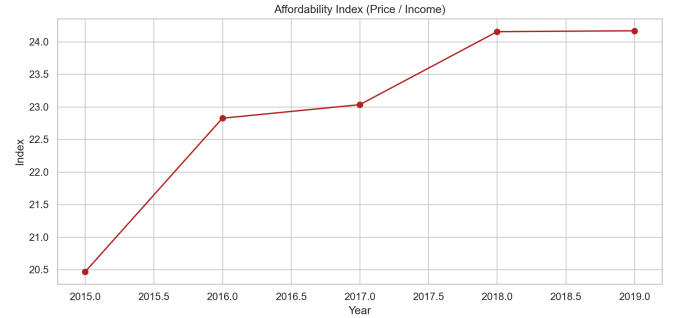


Fig. 2. Affordability index over available overlap years.

The affordability figure uses only years with both price and income.

##### B. Spatial Findings

After improving area labels, bar charts for a chosen year rank localities by mean price; heatmaps across years show whether high-price patches persist. In our outputs, several localities remain at the top of the distribution across multiple years rather than appearing as one-year spikes. That pattern, if robust to better geocoding, would support place-sensitive policy discussion (supply, transport, targeted support) rather than only city-wide averages. Caveat: our areas are derived from text and Eircode prefixes, not official small-area polygons.

##### C. Objective Satisfaction and Evidence Mapping

Table II maps each module-style objective to concrete artifacts: source files, database layers, and outputs. This table is intentional: it ties narrative claims to paths a marker can open in the submitted code bundle without trusting prose alone.

Figures in Section IV use the same yearly and area tables as the dashboard KPIs, so numeric drift would signal a pipeline defect rather than narrative licence.

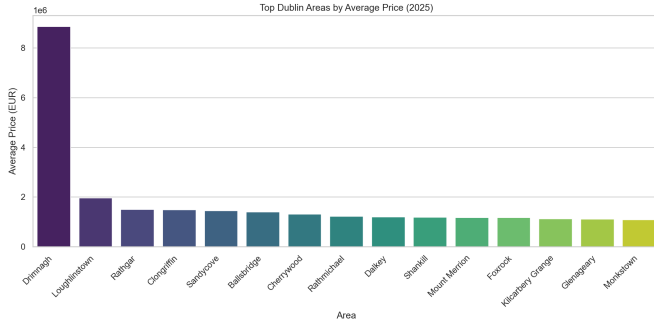


Fig. 3. Top Dublin areas by average price for selected year.

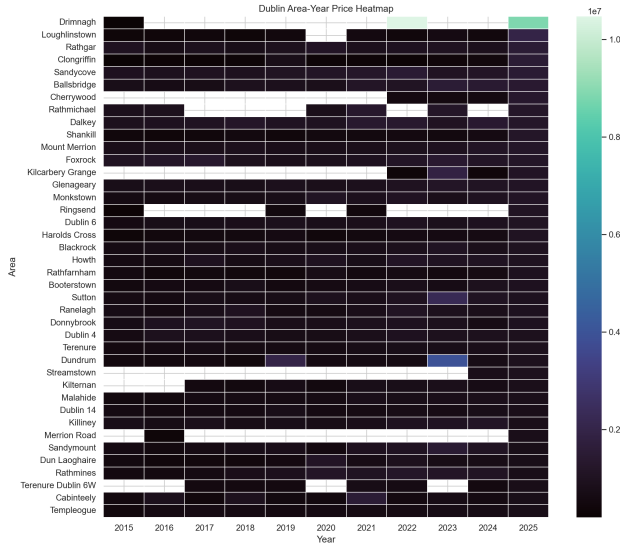


Fig. 4. Area-year heatmap highlighting persistence of high-cost zones.

#### D. Interpretation and Implications

The widening gap between mean prices and available income series is consistent with narratives in which housing quantity responds slowly to demand [3], [4]. We do not estimate structural housing supply elasticities or identify causal effects of policy shocks. Even so, descriptive clarity matters for teaching and for public communication: showing where ratios move and where local means diverge frames questions for specialists to pursue with richer models.

#### E. Critical Reflection

- 1) Income may not cover all property years; ratio claims apply where data overlap.
- 2) Mean price differs from median household experience; no micro household data.
- 3) Area labels are approximate versus official small-area geography.
- 4) Nominal series omit CPI deflation and mortgage conditions; APIs evolve and need maintenance.

#### VI. CONCLUSIONS AND FUTURE WORK

Within years where income data overlap our property aggregates, the price-to-income ratio deteriorates in this project build. Spatial graphics suggest persistent differences between labelled areas, subject to geocoding limitations. Beyond the headline numbers, the main takeaway for readers is methodological: public Irish data *can* be joined reproducibly with disciplined staging, documented cleaning, and dual-database storage that mirrors professional practice (raw document store, relational serve).

**What we learned.** The hardest parts were not plotting lines but agreeing on join keys, handling sparse fields, and refusing to draw charts that extend beyond the data. Those choices mirror real analytics work more closely than fitting a model to a spotless CSV.

**Future work.** We would deflate nominal series with CPI or housing-specific indices; add interest rates or typical mortgage repayments; join to official small-area boundaries; automate runs (e.g. GitHub Actions) with tests that fail when API shapes change; publish checksums of downloads. On tools (LO1): Python integrates the stack; MongoDB fits raw payloads; PostgreSQL fits curated relations; version control remains essential for group accountability—none of these is unique, but together they satisfy the module’s database and pipeline requirements cleanly.

**Process reflection.** Splitting ownership (extract, transform, dashboard) while enforcing shared interfaces reduced merge pain: each stream produced typed outputs the others could import or load, and issues surfaced early in integration meetings rather than on submission day. If we repeated the project, we would add a short data dictionary checked into the repo on day one, because naming debates consumed time that better documentation would have saved.

TABLE II  
MAPPING OF PROJECT OBJECTIVES TO DELIVERABLE EVIDENCE.

| Objective                                         | Where to verify                                            |
|---------------------------------------------------|------------------------------------------------------------|
| Multiple related datasets, semi-structured source | PPR CSV + CSO JSON-stat; parser in <code>extract.py</code> |
| Raw storage in non-relational DB                  | Mongo collections listed in Section III                    |
| Programmatic transform                            | <code>transform.py</code> , <code>main.py</code>           |
| Curated storage in relational DB                  | Postgres tables; <code>load.py</code>                      |
| Analysis + visuals                                | <code>analyze.py</code> , figures, dashboard               |

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